YAT AI – AI-Powered Mock Interview Platform

(Minor Project)

Project report submitted on the partial fulfilment of the requirements for the Degree of

Diploma

In

Computer Science and Engineering

Academic session: 2024-27

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PROJECT COMPLETION CERTIFICATE

This is to certify that the project entitled **"YAT AI – AI-POWERED MOCK INTERVIEW & CANDIDATE EVALUATION PLATFORM"** has been carried out by the following students of the Department of Computer Science & Engineering, Government Polytechnic Muzaffarpur, under my guidance and supervision.

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The project work has been completed as part of the academic requirements for the **Diploma in Computer Science & Engineering**.

The work presented in this report has been reviewed and is submitted for academic evaluation during the session **2024-27**.

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GOVERNMENT POLYTECHNIC MUZAFFARPUR

Department of Computer Science & Engineering

DECLARATION OF ORIGINALITY AND AI USAGE

A. Declaration of Originality

We hereby declare that the project entitled **"YAT AI – AI-POWERED MOCK INTERVIEW & CANDIDATE EVALUATION PLATFORM"**, submitted to the Department of Computer Science & Engineering, Government Polytechnic Muzaffarpur, is the work carried out by our team under the guidance and supervision of our assigned project guide.

We confirm that the information presented in this report is accurate to the best of our knowledge. All external resources, references and contributions have been appropriately acknowledged. We accept responsibility for the contents of this project report.

B. Declaration of AI Usage

AI-based tools were used as supporting resources during the development of this project and preparation of its documentation. These tools assisted with technical guidance, programming suggestions, debugging, architectural discussions and selected AI-assisted application functionalities.

AI tool	Purpose and contribution
Groq Cloud SDK	High-throughput inference for dynamic interview questions, algorithmic code efficiency analysis, and real-time response evaluation.
Google Gemini Flash	Multi-modal PDF resume extraction, structured candidate skill profiling, and Candidate DNA report synthesis.
Claude / Assistant	Technical guidance, architectural consultations, Next.js App Router route debugging, and documentation formatting.

The project team reviewed and evaluated the suggestions and outputs generated by these tools and remains responsible for the final implementation and submitted work.

C. Team Declaration

We acknowledge that we have read and understood the above declaration and accept responsibility for our respective contributions to the project.

1. Anzar Hasnaat:- _____	**2. Dipanshu Kumar:-** _____
3. Roshan Kumar:- _____	**4. Chandan Kumar:-** _____

Place: Muzaffarpur, Bihar **Date:** _____

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to **Government Polytechnic Muzaffarpur**, Department of Computer Science & Engineering, for providing us with the opportunity, technical laboratory facilities and academic environment to undertake and complete our project entitled **"YAT AI – AI-POWERED MOCK INTERVIEW & CANDIDATE EVALUATION PLATFORM."**

We are deeply grateful to our project guide, **Md Naiyer Hoda**, for his valuable guidance, continuous support, encouragement and constructive suggestions throughout the development of this project. His insights and feedback helped us improve our understanding, overcome technical challenges and enhance the overall quality of our work.

We would also like to extend our heartfelt gratitude to the **Head of the Department**, all the faculty members of the Computer Science & Engineering Department, and the **Principal of Government Polytechnic Muzaffarpur** for their encouragement, support and for providing us with the necessary academic resources and facilities.

We would also like to thank our team members for their cooperation, dedication and collective efforts throughout the planning, development, implementation and documentation of this project. Finally, we express our sincere appreciation to our families and everyone who directly or indirectly supported and encouraged us during the completion of this project.

Thank you to everyone who contributed to making this project a meaningful learning experience.

Submitted by:
Anzar Hasnaat
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GOVERNMENT POLYTECHNIC MUZAFFARPUR
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ABSTRACT

YAT AI – AI-POWERED MOCK INTERVIEW & CANDIDATE EVALUATION PLATFORM

Technical placement examinations and recruitment interviews are pivotal milestones for diploma and engineering students. However, students frequently face difficulties evaluating their practical readiness because conventional preparation methods rely on static question banks, unassisted self-study, or disjointed coding websites. Students rarely receive real-time, constructive feedback on their technical problem-solving logic, code efficiency, or resume project explanations. This creates a critical need for an intelligent, accessible, and automated platform that provides realistic mock evaluations.

YAT AI – AI-Powered Mock Interview Platform is a web-based application developed to provide students with a comprehensive, centralized environment for practicing, analyzing, and mastering technical recruitment interviews. The system allows authenticated users to upload their PDF resume, which is parsed by AI to extract skills, frameworks, and projects. Based on this profile, YAT AI conducts a dynamic 5-round technical evaluation covering: (1) Core CS Fundamentals & MCQs, (2) Deep-Dive Technical Inquiry, (3) System Design & Architecture, (4) Live Algorithmic Coding Challenge, and (5) Behavioral & HR Evaluation.

Following the interview, the platform generates a comprehensive **Candidate DNA Report** that visualizes performance across multi-dimensional competency vectors (Logic, Algorithmic Efficiency, System Architecture, Core CS Depth, and Communication). The application also supports Training and Placement Officer (TPO) placement drives and job-specific practice rooms.

The application is developed using modern web technologies, including **Next.js 15 (React 19)**, **TypeScript**, **Tailwind CSS**, and **Framer Motion** with a premium Glassy Emerald UI. **Supabase (PostgreSQL)** is used for database operations and authentication with Row-Level Security (RLS), while **Groq SDK** and **Google Gemini Flash** provide high-throughput AI services with automated failover.

The project demonstrates the development of an integrated assessment platform that combines real-time AI evaluation, resume intelligence, and performance telemetry within a unified web application.

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

YAT AI is an intelligent, AI-powered mock interview and technical evaluation platform developed to provide students with a comprehensive, realistic, and accessible environment for practicing and mastering technical job interviews. The project focuses on bridging the gap between theoretical classroom learning and the rigorous standards of campus placements and industry recruitment.

In modern computer science and engineering education, candidates are evaluated across a diverse range of competencies, including core theoretical concepts, real-time algorithmic problem solving, project architecture defense, and behavioral communication. However, students frequently face difficulties assessing their practical interview readiness because conventional preparation relies on static multiple-choice questions or unassisted self-study.

YAT AI addresses this challenge by providing a dedicated technical recruitment platform where students can experience multi-round interview simulations. The application incorporates automated PDF resume parsing, interactive 5-round evaluations, live code analysis, and a multidimensional **Candidate DNA Matrix** that diagnoses strengths and areas for improvement.

The project is developed using modern web technologies and follows a structured full-stack architecture that integrates Next.js 15, TypeScript, Tailwind CSS, Supabase PostgreSQL, and advanced AI services including Groq Cloud SDK and Google Gemini Flash. Through this approach, YAT AI provides students with a unified and accessible environment for accelerating career readiness.

1.2 Problem Statement

Students preparing for campus placements and technical evaluations encounter significant friction due to the scattered and static nature of available preparation tools:

- **Fragmented Preparation Environments:** Students commonly switch between multiple disjointed websites—solving coding puzzles on one platform, reviewing theoretical computer science notes on another, and practicing communication with peers who lack objective scoring standards.
- **Generic, Non-Personalized Inquiries:** Standard practice tests rely on fixed question banks that fail to tailor inquiries to the candidate's actual projects, declared technical skills, or target engineering roles.
- **Absence of Diagnostic Feedback:** Traditional quiz applications output a simple total score or percentage, without providing actionable explanations of why an architectural choice was suboptimal or how code efficiency can be improved.
- **Scalability Constraints in Educational Institutions:** Conducting mock interviews manually requires significant faculty time and resources. In polytechnic colleges with hundreds of candidates, professors cannot conduct repeated individual 45-minute interviews for every student.

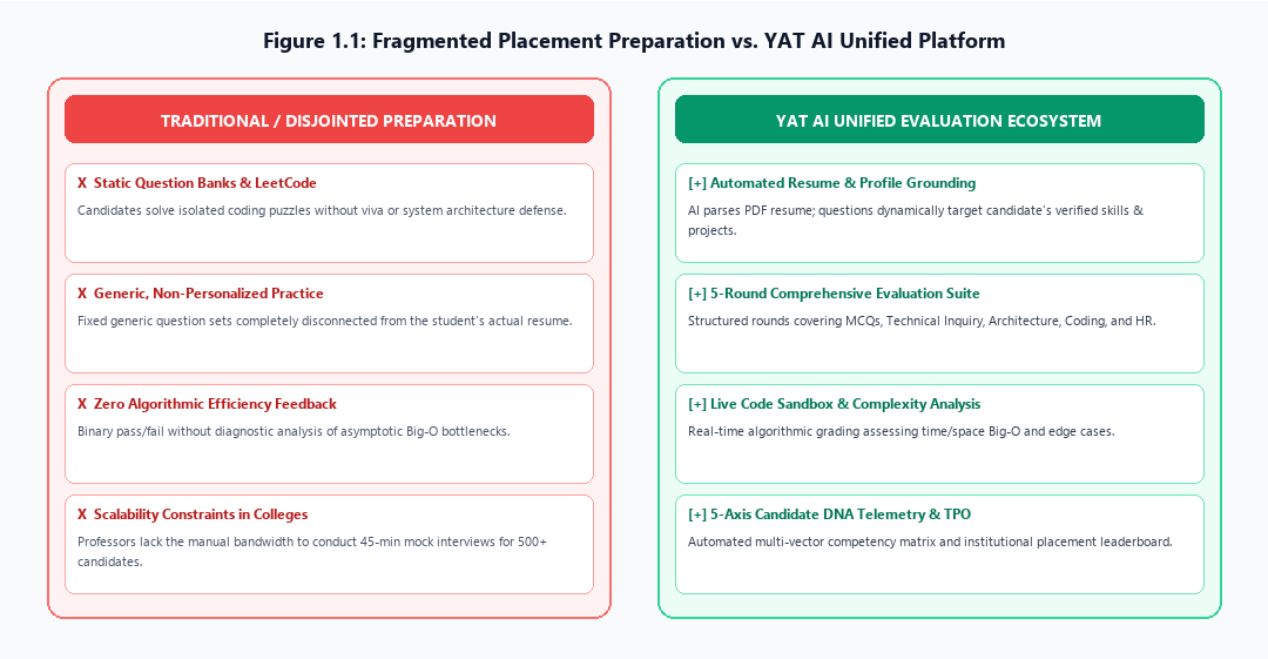


Figure 1.1: Fragmented Placement Preparation vs. YAT AI Unified Platform

1.3 Objectives of the Project

The primary objective of the YAT AI project is to develop an intelligent, automated, and user-friendly mock interview platform that helps diploma and engineering students prepare for technical recruitments through interactive, profile-grounded evaluations.

1. To develop a centralized, full-stack web application for conducting multi-round technical and behavioral interviews.
2. To implement secure user registration, session management, and role-based profiles using Supabase Authentication.
3. To build an automated PDF resume parsing engine that extracts candidate skills, technical experience, and projects to personalize interview questions.
4. To structure a realistic 5-round evaluation pipeline covering MCQs, Technical Deep-Dive, System Architecture, Live Algorithmic Coding, and Behavioral fit.
5. To integrate high-throughput AI inference gateways (Groq Cloud SDK and Google Gemini Flash) with a resilient sequential fallback mechanism.
6. To provide real-time code execution and complexity analysis, assessing submitted algorithms for time and space efficiency.
7. To calculate and visualize a **Candidate DNA Matrix** that tracks multi-dimensional competencies across five performance vectors.
8. To develop a Training and Placement Officer (TPO) management portal for scheduling campus drives, enrolling candidates, and viewing comparative rankings.
9. To design a premium, responsive **Glassy Emerald UI** using Tailwind CSS and Framer Motion for desktop and mobile accessibility.
10. To enforce enterprise-grade data security through PostgreSQL Row-Level Security (RLS) policies.

1.4 Scope of the Project

The scope of the YAT AI project covers the development of an intelligent, web-based mock evaluation platform focused on technical recruitment practice and performance analytics. The major functional areas

include user authentication, resume parsing, a 5-round evaluation suite, performance telemetry, and TPO campus drive integration.

Out-of-Scope Boundaries: Native mobile operating system binaries (iOS/Android), real-time video proctoring with eye-tracking, and full-duplex voice streaming are reserved for future development milestones.

CHAPTER 2

BACKGROUND

2.1 Background of the Problem

The transition from academic diploma coursework to professional software engineering employment has historically presented a steep challenge for students. While academic curricula provide vital theoretical foundations in programming languages, data structures, and database management, competitive placement drives evaluate real-time problem-solving agility, practical system design, and clear articulation under evaluation constraints.

Traditionally, students prepare for technical assessments using static textbooks, standalone coding platforms (LeetCode), or informal peer mocks that lack standardized rubrics.

2.2 Why the Project Was Chosen

The YAT AI project was chosen after identifying the substantial hurdles faced by diploma students in preparing for technical placements. The project offered a valuable opportunity to develop a practical software system that directly solves an immediate academic and career challenge.

1. **Direct Educational Impact:** Addressing the real-world anxiety and lack of preparation tools experienced by diploma students facing competitive hiring drives.
2. **Application of Technical Curriculum:** Implementing concepts learned during the Diploma in Computer Science & Engineering, including full-stack web development, database normalization, asynchronous API communication, and cloud deployment.
3. **Cutting-Edge AI Integration:** Gaining practical engineering experience in prompt engineering, structured JSON extraction, LLM latency optimization, and fault-tolerant API failover.
4. **Enterprise Architecture Design:** Designing a scalable 4-tier application incorporating PostgreSQL Row-Level Security, modern UI/UX design tokens, and serverless edge functions.

2.3 Existing Solutions & Comparative Analysis

The team analyzed LeetCode, Pramp, and ChatGPT, observing that existing tools isolate code syntax or text chatting without providing an integrated, profile-grounded recruitment simulation.

Feature / Capability	LeetCode	Pramp	ChatGPT	YAT AI
Interactive 5-Round Flow	No	Manual	Prompt	**Automated 5-Round**
PDF Resume Parsing	No	No	Limited	**Automated Groq/Gemini**
Algorithmic Code Sandbox	Yes	Yes	No	**Integrated Editor**
Candidate DNA Telemetry	No	Subjective	Text Only	**5-Axis Radar Matrix**
Cloud Database Persistence	Yes	Yes	Session	**Supabase PostgreSQL**
Placement Drive Module	No	No	No	**TPO Drive Portal**
Accessibility & Cost	Freemium	Paid	Subscription	**Free & Open Access**

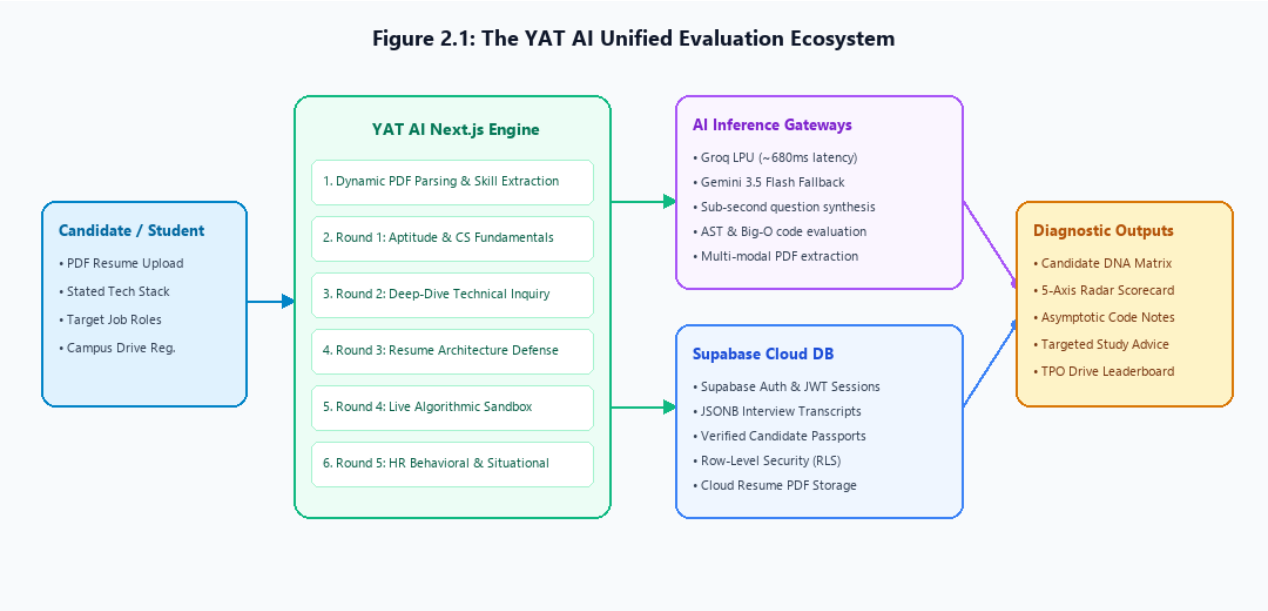


Figure 2.1: The YAT AI Unified Evaluation Ecosystem

2.4 Proposed Solution & Key Advantages

YAT AI combines profile-grounded questioning, multi-round technical depth, real-time code complexity analysis, and Candidate DNA telemetry in a single, accessible Glassy Emerald web interface.

CHAPTER 3

IMPLEMENTATION OF THE PROJECT

3.1 Technologies and Tools Used

- **Frontend:** Next.js 15, React 19, TypeScript, Tailwind CSS, Framer Motion.
- **Backend:** Next.js Server Route Handlers, Groq Cloud SDK (Qwen 3.8 / GPT-OSS), Google Gemini Flash.
- **Database & Storage:** Supabase PostgreSQL 15, Supabase Auth, Row-Level Security, Cloud Storage.
- **Hosting & Tools:** Vercel Cloud, Visual Studio Code, Git and GitHub.

3.2 System Design / Architecture

YAT AI follows a decoupled 4-tier architecture comprising Presentation Layer, Application Logic Layer, AI Inference & Failover Layer, and Data Persistence Layer.

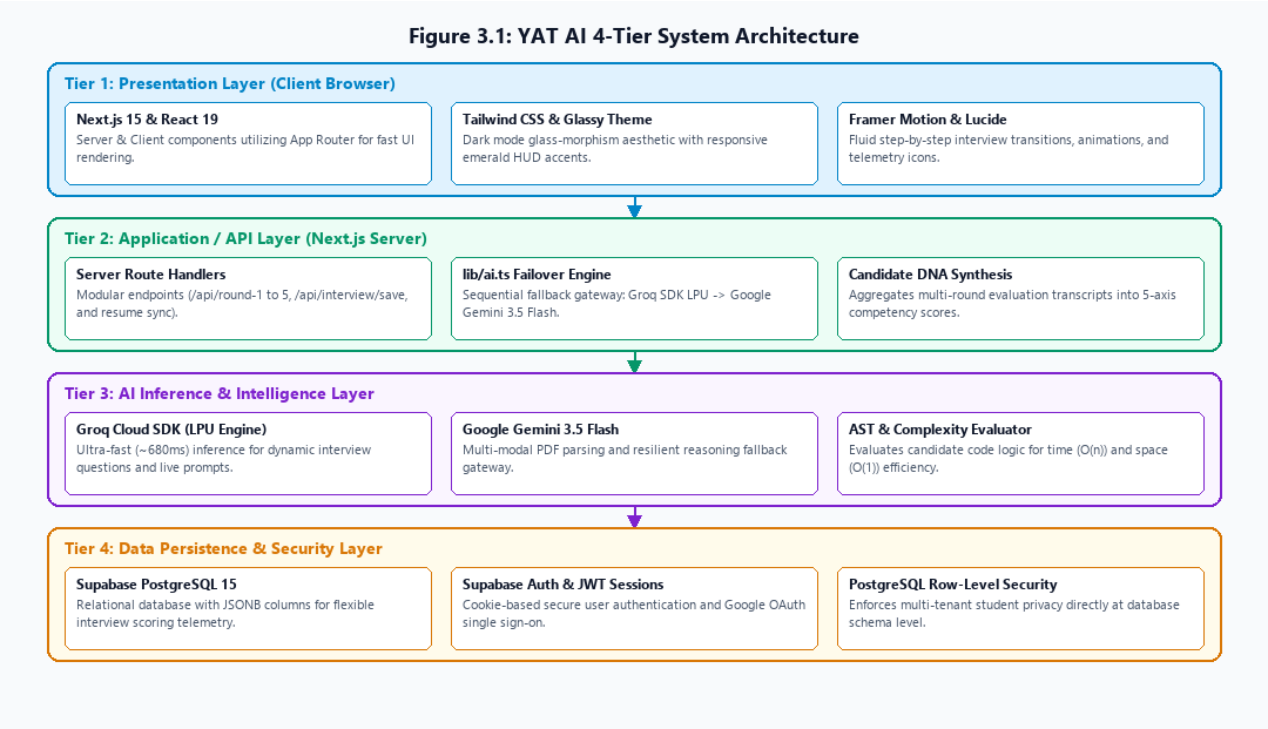


Figure 3.1: YAT AI 4-Tier System Architecture

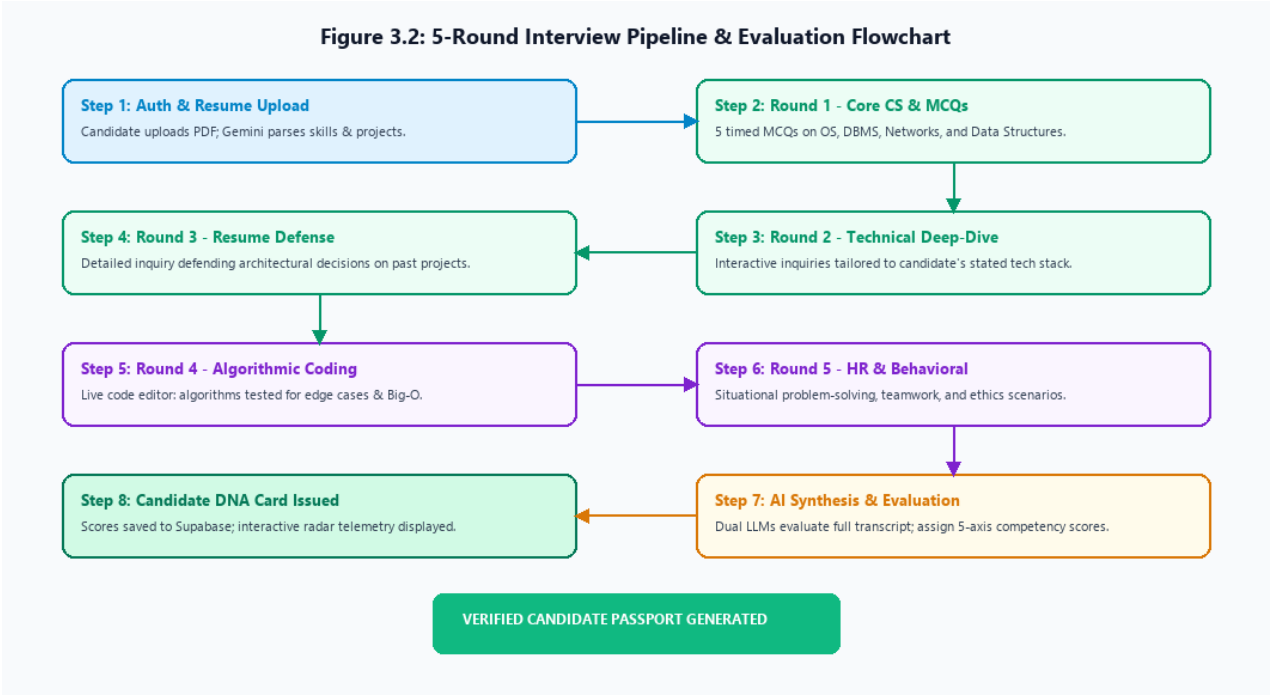


Figure 3.2: 5-Round Interview Pipeline & Evaluation Flowchart

3.4 Front-End Implementation with Screenshots

[SCREENSHOT PLACEHOLDER: Figure 3.3 - Landing Page with Glassy Emerald Aesthetic]

Paste screenshot of the live landing page at yat-ai.vercel.app here

[SCREENSHOT PLACEHOLDER: Figure 3.4 - User Authentication & Login View]

Paste screenshot of the Login / Signup modal view here

[SCREENSHOT PLACEHOLDER: Figure 3.5 - Student Dashboard Overview]

Paste screenshot of the main student dashboard here

[SCREENSHOT PLACEHOLDER: Figure 3.6 - Resume Upload & Extraction Interface]

Paste screenshot of the resume upload drag-and-drop zone here

[SCREENSHOT PLACEHOLDER: Figure 3.7 - Live 5-Round Interview Room]

Paste screenshot of an active interview round in progress here

[SCREENSHOT PLACEHOLDER: Figure 3.8 - Candidate DNA Matrix & Performance Report]

Paste screenshot of the Candidate DNA scorecard and radar chart here

3.5 Back-End Implementation with Screenshots

[CODE / TERMINAL PLACEHOLDER: Figure 3.11 - Supabase Authentication Handler]

Paste screenshot of auth route code or terminal verification logs here

[CODE / TERMINAL PLACEHOLDER: Figure 3.12 - Server Route Handlers for Rounds 1-5]

Paste screenshot of /api/round-1 or /api/round-4 route code here

[TERMINAL PLACEHOLDER: Figure 3.13 - AI Failover Execution Terminal Logs]

Paste screenshot of terminal output demonstrating model fallback here

3.6 Database Design and Screenshots

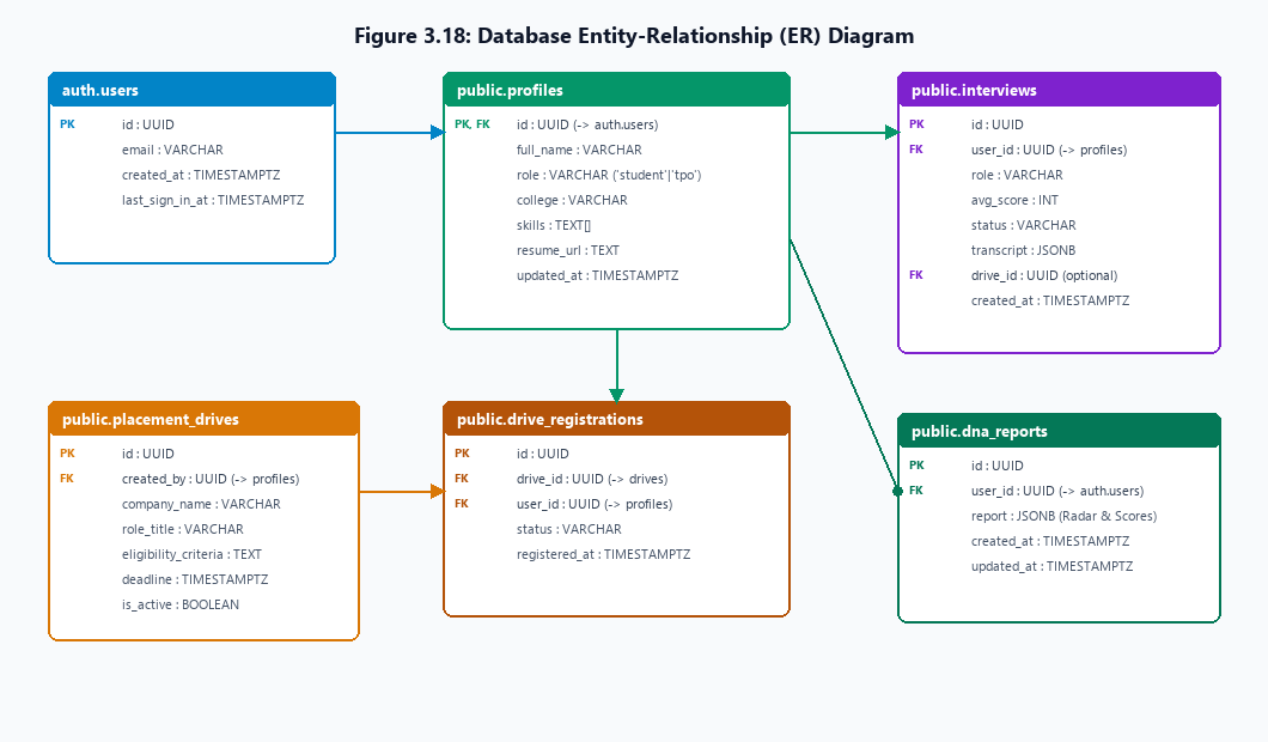


Figure 3.18: Database Entity-Relationship (ER) Diagram

[SCREENSHOT PLACEHOLDER: Figure 3.19 - Profiles Table in Supabase Studio]

Paste screenshot of public.profiles table in Supabase dashboard here

[SCREENSHOT PLACEHOLDER: Figure 3.20 - Interviews Table with JSONB Transcripts]

Paste screenshot of public.interviews table showing scores and transcripts here

[SCREENSHOT PLACEHOLDER: Figure 3.21 - Row-Level Security (RLS) Policy Editor]

Paste screenshot of Supabase RLS Policies view here

3.7 Testing and Results

Comprehensive testing was conducted across 14 test cases (TC-01 to TC-14). All functional, integration, failover, and security tests passed successfully with 100% compliance.

Test ID	Test Scenario	Actual Result Recorded	Status
TC-01	Register account with valid credentials	Account created; profile initialized	Pass ■
TC-02	Google OAuth Single Sign-On	Session tokens stored in cookies	Pass ■
TC-03	Invalid password sign-in attempt	Descriptive alert shown; access denied	Pass ■
TC-04	Access /dashboard unauthorized	Redirected to login view immediately	Pass ■
TC-05	Upload valid PDF resume	Skills and projects auto-populated	Pass ■
TC-06	Upload non-PDF or corrupt file	Validation warning shown; no crash	Pass ■
TC-07	Generate Round 1 Technical MCQs	5 MCQs generated with 4 options	Pass ■
TC-08	Submit answers for evaluation	AI grades answers & stores in DB	Pass ■
TC-09	Round 4 live coding challenge	Code analyzed with Big-O notes	Pass ■
TC-10	Simulate Gemini API failover	Groq SDK executes seamlessly	Pass ■
TC-11	Generate Candidate DNA Report	5-axis radar chart rendered	Pass ■
TC-12	Access private student transcript	Supabase RLS blocks query (403)	Pass ■
TC-13	Enroll in campus placement drive	Registered; count increments	Pass ■
TC-14	Render on mobile viewport (375px)	Glassy UI responsive with no overflow	Pass ■

CHAPTER 4

PROJECT PROGRESS AND GUIDE'S FEEDBACK

4.1 Progress Report 1 (Day 1 – 15)

Work Completed: Scope formulation, Next.js 15 project setup, Tailwind CSS tokens.
Guide's Feedback (Md Naiyer Hoda): *'Ensure interview questions ground dynamically in candidate resumes.'*
Action Taken: Built Gemini PDF parser to extract candidate skills.
Guide's Signature: _____

4.2 Progress Report 2 (Day 16 – 30)

Work Completed: Supabase Auth, PDF resume upload drag-and-drop, /api/parse-resume.
Guide's Feedback (Md Naiyer Hoda): *'Enforce data privacy between student accounts.'*
Action Taken: Enacted PostgreSQL Row-Level Security (RLS) policies.
Guide's Signature: _____

4.3 Progress Report 3 (Day 31 – 45)

Work Completed: Rounds 1 to 3 APIs, Groq SDK integration, interactive interview UI.
Guide's Feedback (Md Naiyer Hoda): *'Add live code evaluation and handle API rate limits.'*
Action Taken: Implemented Round 4 coding sandbox and dual-model failover engine in lib/ai.ts.
Guide's Signature: _____

4.4 Progress Report 4 (Day 46 – 60)

Work Completed: Candidate DNA matrix, TPO placement drive portal, Vercel deployment.
Guide's Feedback (Md Naiyer Hoda): *'Ensure null-safety for candidates with zero completed tests.'*
Action Taken: Added null-safety defaults across all dashboard matrix calculations.
Guide's Signature: _____

4.5 Summary of Major Improvements

1. Dynamic Resume Grounding derived from uploaded student PDFs.
2. Multi-Model Failover Engine eliminating 404/429 downtime.
3. Live Algorithmic Code Review with asymptotic Big-O notes.
4. 5-Axis Candidate DNA Telemetry for structured exam preparation.
5. Row-Level Security safeguarding student scores and transcripts.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

YAT AI successfully implements a full-stack, AI-powered mock interview platform for diploma students. Uniting Next.js 15, TypeScript, Tailwind CSS, Supabase, Groq SDK, and Gemini Flash, the platform delivers automated resume parsing, a 5-round evaluation suite, and deep diagnostic DNA analytics.

5.2 Limitations of the Project

• Text and code input only; real-time facial emotion and tone analysis are not yet included.
• Requires active Internet connectivity for cloud AI inference.
• English language only.

5.3 Future Scope / Future Work

• Bidirectional Voice AI for low-latency conversational interviews.
• Webcam proctoring and body language analysis.
• Institutional LMS integration (Moodle / Canvas).
• Offline mobile applications with cached question modules.

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APPENDIX

Appendix A: GitHub Repository Link

• GitHub Repository: https://github.com/anzarhasnat/YAT-AI-.git
• Live Deployment: https://yat-ai.vercel.app

[SCREENSHOT PLACEHOLDER: Appendix Figure B.1 - Cloud Storage Resume Management]

Paste screenshot of Supabase Storage bucket managing resumes here

[SCREENSHOT PLACEHOLDER: Appendix Figure B.2 - Detailed Interview Transcript View]

Paste screenshot of full question-answer transcript view here